# UNIVERSITY OF LONDON
## INTERNATIONAL PROGRAMMES

## BSc Computer Science and Related Subjects

# CM3070 PROJECT
# FINAL PROJECT REPORT

< Predicting if Videos on Social Media will go Viral>

Author: Siah Cheng Yun
Student Number : 210501959
Template: CM3005 Data Science
https://github.com/Cysiah24/CM3070-Final-Project

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# CHAPTER 1: INTRODUCTION (664 words)

## 1.1 Project Idea

The explosive growth of short-form video content across platforms such as YouTube, TikTok, and Instagram Reels has created a landscape in which millions of videos compete for audience attention every day. Among these, only a small fraction achieves viral status, reaching audiences that far exceed the creator's existing follower base. Understanding what drives this phenomenon has significant implications for content creators, digital marketers, platform engineers, and researchers studying online information diffusion. This project proposes the development of a machine learning system capable of predicting whether a social media video will go viral based on its metadata and early engagement signals.

This project follows the Data Science Template, structured around a predictive modelling pipeline. The workflow encompasses data acquisition, exploratory data analysis, feature engineering, model training and evaluation, and the interpretation of results. The primary dataset used is the publicly available YouTube Trending Video Dataset from Kaggle, which provides records of trending videos across multiple countries, including metadata such as titles, tags, view counts, likes, comments, category information, and channel statistics.

## 1.2 Research Questions and Aims

This project aims to address the following research questions:

**1:** Can video metadata features (such as title characteristics, tags, category, publish time, and channel attributes) be used to predict whether a video will achieve viral levels of engagement?

**2:** Which features are the strongest predictors of virality, and do they differ across content categories or geographic regions?

**3:** How do early engagement signals (e.g. the ratio of likes to views within the first observation window) improve prediction accuracy compared to metadata-only models?

**4:** What constitutes a robust and generalisable definition of "virality" that accounts for differences in creator audience size and content category norms?

The overarching aim is to build an interpretable, reproducible predictive model that not only achieves strong classification performance but also yields actionable insights into the mechanics of viral content.

## 1.3 Deliverables and Objectives

The project is organized around four core deliverables; each tied directly to a specific objective:

**Deliverable 1 – Exploratory Data Analysis Report:** A comprehensive analysis of the YouTube Trending Video Dataset, including distributions of engagement metrics, temporal patterns in publishing and trending behavior, category-level comparisons, and correlation analyses among features. This deliverable addresses the foundational understanding required to inform feature selection and model design.

**Deliverable 2 – Trained Predictive Model and Evaluation Report:** A comparative evaluation of multiple classification algorithms (including Logistic Regression, Random Forest, Gradient Boosted Trees, and a neural network baseline) trained on the engineered features. The evaluation will use precision, recall, F1-score, and ROC-AUC as primary metrics, with particular attention to performance under class imbalance via stratified cross-validation and SMOTE oversampling.

**Deliverable 3 – Feature Importance and Interpretability Analysis:** An analysis of model outputs using permutation importance and built-in feature importance to identify which features contribute most to virality predictions. This deliverable provides the interpretive layer that transforms a black-box classifier into actionable insights for content strategy.

## 1.4 Justification

The objectives and deliverables described above are structured to progressively build toward the project's overarching aims. The exploratory analysis (Deliverable 1) establishes the empirical foundation by revealing patterns and anomalies in the data that directly inform the feature engineering process.

The predictive model (Deliverable 2) serves as the primary technical contribution, testing whether virality can be anticipated from observable features. By comparing metadata-only models against models that also incorporate early engagement ratios, the evaluation explicitly isolates the incremental predictive value of each feature category. Finally, the interpretability analysis (Deliverable 3) ensures the project delivers practical value beyond classification accuracy, enabling stakeholders to understand which levers are most associated with viral outcomes.

Together, these deliverables form a coherent pipeline from raw data to actionable insight. The project contributes to the growing body of research on social media content performance prediction, with the added rigor of interpretable machine learning methods and a carefully constructed virality metric. The use of a publicly available, continuously updated dataset also ensures reproducibility and provides a foundation for future extensions, including cross-platform analysis and real-time prediction deployment.

# CHAPTER 2: LITERATURE REVIEW (1634 words)

## 2.1 The Psychology of viral Content

Any attempt to predict virality computationally must first engage with the question of why content goes viral in the first place. The most influential study in this area is Berger and Milkman's (2012) investigation of what drives online sharing, published in the Journal of Marketing Research. Analyzing every article published on the New York

Times website over a three-month period, the authors found that virality is not random: content that evokes high-arousal emotions such as awe, anger, or anxiety is significantly more likely to be shared, whereas content eliciting low-arousal emotions such as sadness suppresses sharing. Positive content was shared more than negative content overall, but the relationship was mediated by physiological arousal rather than valence alone. These findings held even after controlling for an article's practical usefulness, surprise value, and the prominence with which it was featured on the homepage.

Berger and Milkman's work establishes that virality is partially predictable from content characteristics, providing the theoretical rationale for building a machine learning classifier. Their focus on emotional arousal motivates the inclusion of sentiment analysis features in the feature engineering pipeline: if emotional tone can be computationally extracted from video titles, descriptions, and tags, it may serve as a meaningful predictor. However, a key gap in their work is that it focused on text-based news articles rather than video content, and it relied on regression analysis rather than modern machine learning methods. This project addresses both gaps by applying classification algorithms to video metadata and engagement data, testing whether the psychological principles identified by Berger and Milkman translate into quantifiable predictive features for video content.

## 2.2 Similar Projects: Predicting Video Popularity

Several research teams have directly tackled the problem of predicting video popularity using computational methods. Pinto, Almeida, and Gonçalves (2013) presented two models for forecasting YouTube video popularity based on early view patterns, published at the ACM International Conference on Web Search and Data Mining (WSDM). They proposed a multivariate linear regression model that incorporates historical viewing trajectory information from the first few days after upload to predict future view counts. Their key insight was that videos follow distinct temporal popularity patterns, and that the shape of the early viewing curve carries predictive information. This work directly

motivates the inclusion of early engagement signals as features in the present project, and specifically the decision to compare metadata only models against models that also incorporate early engagement ratios.

Berkovsky et al. (2015) extended this line of research by examining the cross-platform dynamics between YouTube and Twitter, published at the ACM Conference on Information and Knowledge Management (CIKM). They collected a corpus of tweets containing YouTube links alongside the corresponding video metadata and found that YouTube engagement features, particularly user interaction metrics, had strong predictive power for virality. A key finding was that cross-system prediction was feasible: features extracted solely from a video-sharing service could predict virality on a social network and vice versa. Their framework achieved high predictive accuracy with relatively small amounts of training data. This study is important to the present project because it demonstrates that metadata-level features carry genuine predictive signal for virality, validating the use of the YouTube Trending Video Dataset's metadata fields as model inputs.

Triantafyllou et al. (2022), in their study published at ACM CHI, examined TikTok short videos and identified specific visual and structural indicators of virality. Using a mixed-methods approach on 400 labelled TikTok videos, they found that while creator follower count was the most powerful predictor, close-up and medium-shot video framing, video lifespan, presence of on-screen text, and the point of view used in filming also played significant roles. This work is notable because it demonstrates the importance of content-level features beyond simple engagement counts and suggests that even metadata proxies for content characteristics (such as video duration, which may correlate with shot type and editing style) can be informative. The present project does not have access to visual frame-level features given the metadata-only nature of the YouTube Trending dataset, but the findings motivate the engineering of as many content-proxy features as possible from the available metadata.

More recently, Javed et al. (2025) proposed a framework combining engagement metrics with Natural Language Processing-derived sentiment scores from user comments to classify videos into popularity tiers. Published in Engineering Reports, they evaluated four machine learning models: Random Forest, Stochastic Gradient Descent, Gradient Boosting, and XGBoost. They found that XGBoost achieved the highest accuracy at 84.3%, outperforming baselines by up to 20%. This study directly validates the choice of gradient boosted tree models for the present project and demonstrates that integrating textual sentiment features meaningfully improves classification performance over purely numerical engagement features.

## 2.3 Techniques and Methods

**Handling Class Imbalance with SMOTE.** A fundamental challenge in virality prediction is the extreme class imbalance inherent in the problem: by definition, viral videos are rare events. Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE) in the Journal of Artificial Intelligence Research, which generates synthetic examples of the minority class by interpolating between existing minority instances and their nearest neighbors, rather than simply duplicating them. The authors demonstrated that combining SMOTE with majority-class under-sampling achieved better classifier performance in ROC space than under-sampling alone, across multiple classifiers including C4.5 decision trees and Naive Bayes. With over 28,000 citations, SMOTE has become the standard approach for addressing class imbalance in machine learning. This technique is directly applicable to the present project, where the proportion of "viral" videos (however defined) will constitute a small fraction of the dataset. The project will evaluate classification performance both with and without SMOTE to quantify its impact on recall and F1-score for the minority (viral) class.

**Model Interpretability with SHAP.** Predicting virality is only useful if the model's decisions can be understood and acted upon. Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations) at the Conference on Neural Information Processing Systems (NeurIPS), presenting a unified framework for interpreting machine learning

predictions. SHAP assigns each feature an importance value for a particular prediction based on Shapley values from cooperative game theory, which represent the unique, provably fair allocation of contribution among features. The framework unifies six previously disconnected interpretability methods and provides both local explanations (why a specific prediction was made) and global explanations (which features matter most overall). The paper demonstrated that SHAP values show improved consistency with human intuition compared to prior approaches, and introduced computationally efficient variants such as TreeSHAP for tree-based models.

**Gradient Boosted Decision Trees.** The choice of classification algorithm is informed by a substantial body of evidence showing the effectiveness of ensemble tree methods for tabular prediction tasks. Gradient Boosted Decision Trees (GBDTs), as formalized by Friedman (2001) and popularized through implementations such as XGBoost (Chen and Guestrin, 2016) and LightGBM (Ke et al., 2017), have consistently achieved state-of-the-art performance on structured data. In the specific context of video popularity prediction, Javed et al. (2025) found XGBoost to be the top-performing classifier, and Nagashree et al. (2025) similarly employed machine learning on YouTube engagement data with strong results. GBDTs are well-suited to the present project because they handle mixed feature types (numerical and categorical) naturally, are robust to feature scale differences, and integrate seamlessly with TreeSHAP for interpretability. The project will benchmark GBDT performance against Logistic Regression (as an interpretable baseline), Random Forest (as an alternative ensemble), and a simple neural network (to assess whether non-linear feature interactions improve performance).

## 2.4 Cascade and Diffusion Models

An alternative paradigm for predicting virality focuses on modelling the diffusion process itself rather than predicting from static features. Kong et al. (2018) built the HIPie system on top of the Hawkes Intensity Process (HIP), a self-exciting point process model that captures the endogenous amplification of popularity (where views breed more views) and the sensitivity to external promotion. Published at The Web Conference (WWW), the tool

enables users to visualize a video's position in an endogenous-exogenous space that indicates its viral potential and to simulate the effect of different promotion strategies. Similarly, the ViralGCN framework presented by researchers in *Mathematics* (2023) used a temporal-spatial graph convolutional network to model cascade dynamics, identifying four distinct types of users in the diffusion process (broadcasters, influential disseminators, active responders, and secondary responders) and finding that both broadcast diffusion and structural virality contribute to a video going viral.

These process-based models are complementary to the feature-based approach adopted in the present project. While they offer richer theoretical accounts of how content spreads, they require access to detailed cascade data (individual sharing events with timestamps) that is not available in the YouTube Trending dataset. The present project therefore focuses on feature-based prediction as the more practical and widely applicable approach, while acknowledging that future work could integrate cascade dynamics if per-video sharing logs were obtainable.

## 2.5 Identified Gaps and Project Contribution

The literature review reveals several gaps that the present project is positioned to address. First, while the psychology literature (Berger and Milkman, 2012) establishes that emotional content drives sharing, the translation of these findings into computable features for video metadata has been limited; the present project explicitly engineer's sentiment and textual features from titles and tags to test this connection. Second, most existing video prediction studies either focus on raw view count regression (Pinto et al., 2013) or binary popularity classification without carefully defining the virality threshold; this project introduces a normalized virality metric that accounts for creator audience size and content category norms. Third, while studies such as Javed et al. (2025) have demonstrated strong model performance, few have combined high-performing classifiers with rigorous interpretability analysis; this project pairs GBDT models with permutation importance to deliver actionable insights alongside accurate predictions. Finally, the explicit comparison of metadata-only models against models incorporating early

engagement signals provides a structured evaluation of feature category value that is largely absent from the existing literature.

# CHAPTER 3: PROJECT DESIGN (1487 words)

## 3.1 Domain and Target Users

This project sits at the intersection of social media analytics and predictive data science. The domain encompasses the study of online video content performance on user-generated content platforms, with YouTube serving as the primary case study. The domain is characterized by extremely high data volumes, a power-law distribution of engagement (where a tiny fraction of content captures most views), and a complex interplay between content quality, algorithmic recommendation, creator influence, and audience behavior.

The project serves three primary user groups. Content creators and influencers benefit the most by understanding which metadata-level features correlate with virality enables them to make data-informed decisions about titles, tags, publishing schedules, and category selection. Digital marketing professionals and brand strategists can use the model's interpretability outputs to guide investment in video campaigns, optimizing content strategies based on empirical evidence rather than intuition. Platform researchers and data scientists studying recommendation systems and content dynamics benefit from the feature importance analysis, which reveals which observable signals are most associated with algorithmic amplification and audience engagement. A secondary beneficiary group includes academic researchers in computational social science who study information diffusion and online attention dynamics.

The needs of these users directly inform the design of the project. Content creators require not just a prediction but an understanding of why certain features matter. Marketing professionals need confidence in model reliability across different content

categories and regions, motivating the cross-category evaluation. Researchers require reproducibility, driving the decision to use a publicly available dataset and document the full pipeline.

## 3.2 Design Choices and Justification

**Dataset selection**

The dataset was chosen over alternatives for several reasons: it provides metadata across multiple countries enabling geographic comparison; it is continuously maintained, ensuring the data reflects current platform dynamics; it includes both engagement metrics (views, likes, comments) and content metadata (title, tags, category, channel); and it has an active community of users with existing notebooks that can inform preprocessing decisions. The decision to use an existing dataset rather than collecting data via the YouTube API avoids rate-limiting constraints and ensures immediate reproducibility.

**Virality definition**

Rather than using raw view count as the target variable which conflates virality with pre-existing audience size, the project defines virality using a normalized engagement ratio: the ratio of total engagement (views + likes + comments) to the channel's subscriber counts at the time of trending. Videos in the top 5th percentile of this ratio within their content category are labelled as "viral." This threshold was chosen because it produces a meaningfully rare positive class that aligns with the intuitive notion that virality is an exceptional outcome, while still providing sufficient positive samples for model training. The category-level normalization accounts for the fact that, for example, music videos routinely achieve higher raw view counts than educational content without being "viral" in a meaningful sense.

**Feature engineering strategy**

Features are organized into two tiers. Tier 1 (metadata-only) includes features derivable before or at the time of publishing: title length, title word count, title sentiment score,

number of tags, tag diversity, publish hour and day-of-week (cyclically encoded), video category, and channel age. Tier 2 (early engagement) adds features observable within the first trending observation: likes-to-views ratio, comments-to-views ratio, and the like-to-dislike ratio. This two-tier design enables a controlled comparison that directly addresses Research Question 3 by isolating the incremental value of early engagement signals.

**Model selection.** Four classifiers were selected to provide a spectrum of complexity and interpretability. Logistic Regression serves as the interpretable baseline, offering direct coefficient interpretation. Random Forest provides an ensemble benchmark with built-in feature importance. XGBoost is the primary candidate based on its demonstrated effectiveness in similar tasks (Javed et al., 2025). A shallow neural network tests whether non-linear feature interactions improve performance. All models are evaluated under identical cross-validation protocols to ensure fair comparison.

## 3.3 Project Architecture

The project follows a standard supervised machine learning pipeline, structured into five sequential stages with clearly defined inputs and outputs:

**Stage 1: Data Acquisition and Preprocessing.**

The raw CSV files from the YouTube Trending dataset are loaded, deduplicated as videos can appear on trending lists across multiple days, and cleaned. Missing values in tags and descriptions are handled via imputation (empty string for text, median for numerical). Date fields are parsed and standardised. The output is a clean, unified data frame stored in Parquet format for efficient downstream processing.

**Stage 2: Exploratory Data Analysis**

The cleaned data is profiled through distribution analysis such as engagement metrics, temporal patterns, category breakdowns, correlation analysis, and outlier detection. Visualisations are generated using Matplotlib and Seaborn. The output is the EDA Report (Deliverable 1) and a set of insights that inform feature selection.

**Stage 3: Feature Engineering and Target Construction**

Derived features are computed from raw fields (sentiment scores, temporal encodings, engagement ratios). The virality label is computed using the normalised engagement ratio with category-level percentile thresholds. Features are scaled (StandardScaler for numerical, one-hot encoding for categorical). The output is the Feature-Engineered Dataset (Deliverable 2), split into training (80%) and test (20%) sets with stratified sampling to preserve class proportions.

**Stage 4: Model Training and Evaluation**

Each classifier is trained using 5-fold stratified cross-validation on the training set. Hyperparameter tuning is performed via Randomized Search CV for tree-based models and grid search for logistic regression. SMOTE is applied within each cross-validation fold after splitting to prevent data leakage. Models are evaluated on the held-out test set using precision, recall, F1-score, and ROC-AUC. The output is the Model Evaluation Report (Deliverable 2).

**Stage 5: Interpretability Analysis**

Permutation importance and built-in impurity-based importance are computed for the best-performing model to identify which features contribute most to predictions. Permutation importance works by shuffling each feature's values and measuring the resulting drop in F1-score, providing a model-agnostic measure of feature contribution. Dependence plots are generated for the top three features to visualise non-linear relationships between feature values and predicted viral probability. The output is the Feature Importance and Interpretability Analysis (Deliverable 3)

## 3.4 Technologies and Methods

The project is implemented entirely in Python, chosen for its dominant position in the data science ecosystem and the maturity of its machine learning libraries. The core technology stack is as follows:

| Component | Technology | Justification |
|---|---|---|
| Data handling | Pandas, NumPy | Industry-standard libraries for tabular data manipulation and numerical computation. |
| Visualisation | Matplotlib, Seaborn | Publication-quality static visualisations for EDA and results presentation. |
| Sentiment | VADER (vaderSentiment) | Lexicon-based sentiment suited to short informal text like video titles. |
| ML framework | Scikit-learn | Unified API for preprocessing, cross-validation, model training, and evaluation metrics. |
| Class imbalance | Imbalanced-learn (SMOTE) | Integrates with Scikit-learn pipelines. Supports within-fold application to prevent data leakage. |
| Interpretability | Permutation importance | Model-agnostic feature importance built into Scikit-learn. |
| Environment | Jupyter Notebooks | Combines code, output, and narrative for reproducible, documented analysis workflows. |

The choice of VADER over transformer-based sentiment models (e.g. BERT) is deliberate: video titles and tags are short, informal text where VADER's lexicon-based approach performs comparably to deep learning models while being orders of magnitude faster.

## 3.5 Evaluation Plan

The project will be evaluated using a held-out test set approach. The dataset is split 80/20 with stratified sampling so the test set preserves the original class distribution. SMOTE is applied only to the training set to ensure the test set represents real-world conditions. F1-score for the viral class is the primary metric rather than accuracy, because with 85% non-viral videos a model predicting 'not viral' for everything would achieve 85% accuracy while being useless. ROC-AUC is used as a complementary metric measuring discrimination across all thresholds.

Three controlled comparisons are designed to isolate specific effects. The four-model comparison holds features constant while varying the algorithm. The tier comparison holds the algorithm constant while varying the feature set. The SMOTE comparison holds everything constant except oversampling. Each comparison changes exactly one variable.

The evaluation targets are:

| Criterion | Target | Rationale |
|---|---|---|
| F1-score (viral class) | ≥ 0.35 | Meaningful improvement over random baseline on imbalanced data. |
| ROC-AUC | ≥ 0.70 | Demonstrates genuine discriminative ability. |
| Tier 2 > Tier 1 | Improvement | Engagement features add measurable value. |
| SMOTE impact | Positive | Oversampling improves minority class detection. |

| | | |
|---|---|---|
| CV stability (σ) | ≤ 0.05 | Results not sensitive to data split. |
| Reproducibility | Verified | Full pipeline runs end-to-end from raw data. |

## 3.6 Work Plan

The project spans 12 weeks and is organised into the phases shown in the Gantt chart below. Each task is designed to be completable within a two-week window. Deliverable milestones are highlighted to ensure regular, incremental progress. Shaded cells indicate active work periods.

| Task | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | W11 | W12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Data acquisition & cleaning | | | | | | | | | | | | |
| Exploratory data analysis | | | | | | | | | | | | |
| EDA report (Deliverable 1) | | | | | | | | | | | | |
| Feature engineering | | | | | | | | | | | | |
| Model training & tuning | | | | | | | | | | | | |
| Model comparison & evaluation | | | | | | | | | | | | |
| Model report (Deliverable 2) | | | | | | | | | | | | |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Interpretability report (Deliv. 3) | | | | | | | | | | | | |
| Final report writing | | | | | | | | | | | | |
| Review, refinement & submission | | | | | | | | | | | | |

**Key Milestones:**

**Week 4:** EDA Report (Deliverable 1) complete. Feature engineering design finalised.

**Week 6:** Model training begins.

**Week 9:** Model Evaluation Report (Deliverable 2) complete. Best model selected.

**Week 10:** Interpretability Analysis (Deliverable 3) complete.

**Week 12:** Final report submitted.

# CHAPTER 4: IMPLEMENTATION (1028 words)

This chapter describes the major algorithms and techniques used in the virality prediction pipeline, explains the most important parts of the code, and presents visual representations of the results.

## 4.1 Data Pipeline and Preprocessing Algorithm

The raw dataset contains 120,746 rows across three CSV files. Since videos can appear on the trending page for multiple consecutive days, the deduplication algorithm sorts by trending date and keeps only the latest observation per video per country, retaining the highest engagement values. The code also loads JSON category mappings, resolves a timezone mismatch between the tz-aware publish timestamps and tz-naive trending dates, and removes errored videos.

```
# --- Parse dates ---
# trending_date format is YY.DD.MM in this dataset
df['trending_date'] = pd.to_datetime(df['trending_date'], format='%y.%d.%m', errors='coerce')

# publish_time is ISO format
df['publish_time'] = pd.to_datetime(df['publish_time'], errors='coerce')

# --- Deduplication ---
# Videos appear on trending for multiple days; keep the latest observation
df = df.sort_values('trending_date')
df = df.drop_duplicates(subset=['video_id', 'country'], keep='last')
print(f"  After deduplication: {len(df)} rows")
```

After cleaning, 34,025 unique videos remain. The virality label is computed using a normalized engagement score: (views + likes + comments) divided by the channel's median view count, which serves as a proxy for subscriber count since the dataset lacks that field. Videos in the top 15th percentile of this score within their category are labelled viral, producing 5,111 viral (15%) and 28,914 non-viral (85%). Figure 1 shows the resulting distribution.

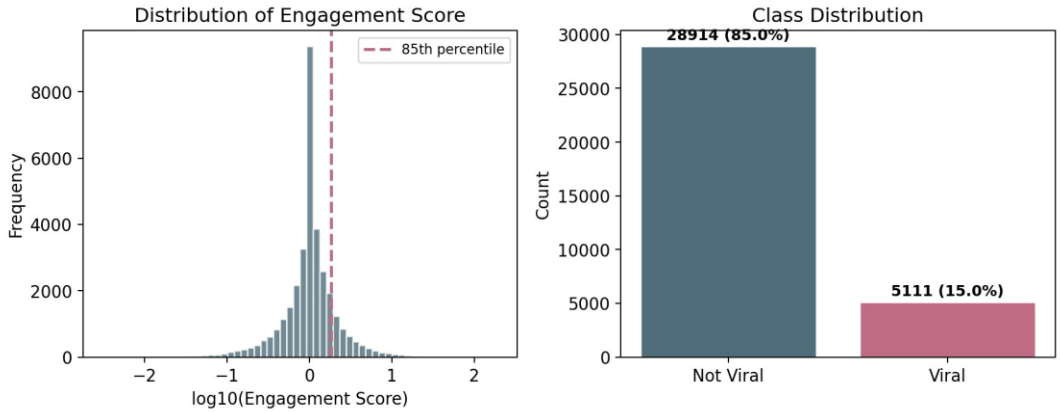


*Figure 1: Engagement score distribution and resulting class balance.*

## 4.2 Exploratory Data Analysis

The EDA stage uses Matplotlib and Seaborn to generate five visualizations that directly inform the modelling decisions. Figure 2 shows Entertainment dominates by volume while the "Movies" category has near-100% viral rate due to tiny sample size, confirming this is an outlier. Other categories cluster around 15–20%, validating the category-level normalization.

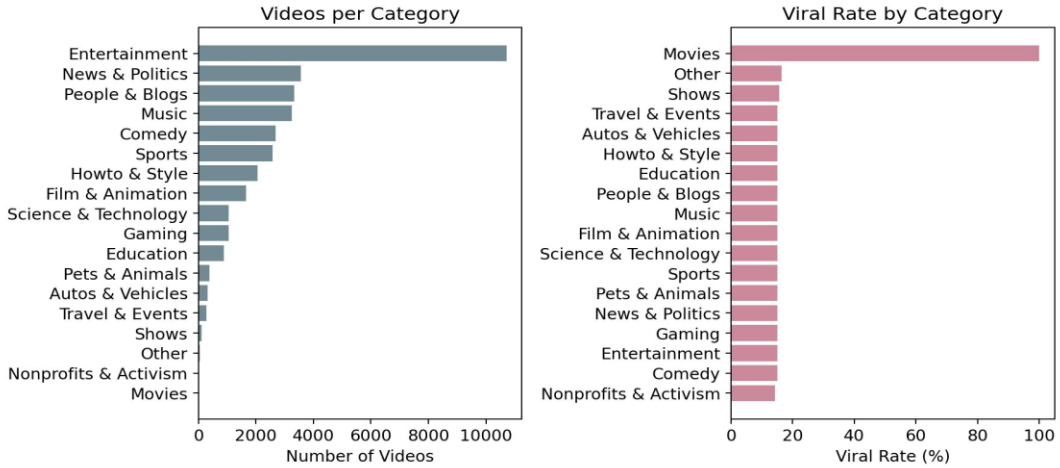


*Figure 2: Videos per category and viral rate by category.*

Figure 3 reveals that raw engagement metrics are heavily correlated (likes-views r=0.78, likes-dislikes r=0.44, likes-comments r=0.76), which motivated the decision to use

engagement ratios instead of raw counts to avoid multicollinearity. Title sentiment shows weak but non-zero correlations.

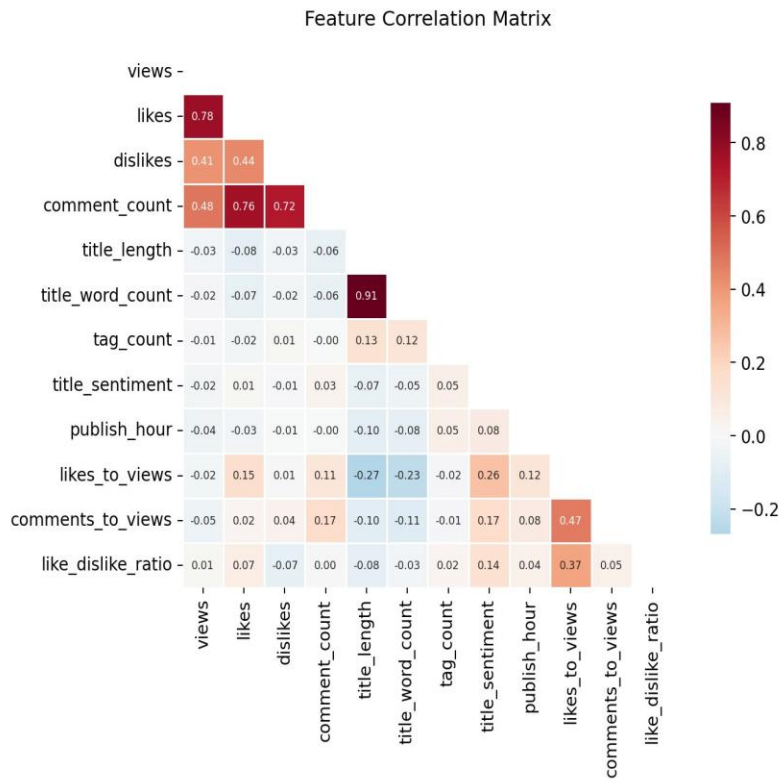


*Figure 3: Feature correlation matrix.*

Figure 4 shows that videos published between 3:00–6:00 UTC have the highest viral rates (~21%), likely corresponding to US prime time. Day-of-week effects are flat.

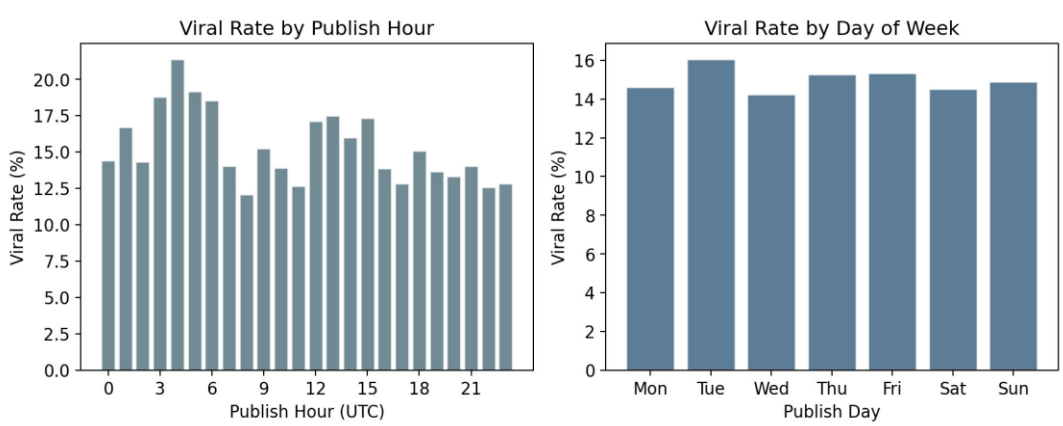


*Figure 4: Viral rate by publish hour and day of week.*

Figure 5 shows the distributions of key features are nearly identical between viral and non-viral videos, which is the central challenge for the classification models.

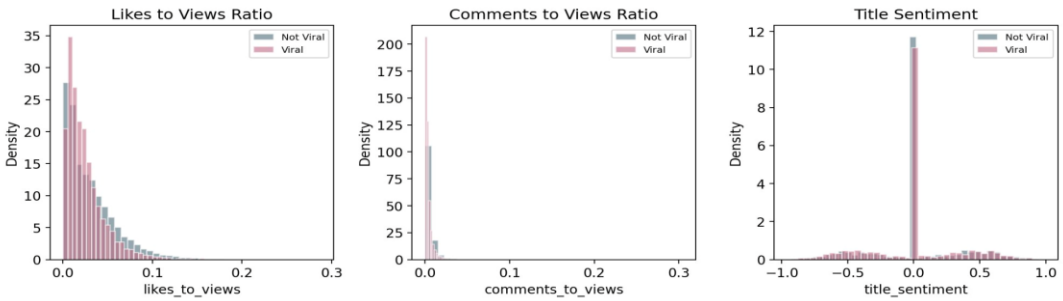


*Figure 5: Heavy overlap in feature distributions between classes.*

## 4.3 Feature Engineering Techniques

Three key transformation techniques are applied. Cyclical encoding represents publish hour and day of week as sine/cosine pairs so that the model understands 11pm and midnight are close together, not far apart. Log transformation reduces the extreme right skew of views (from 24.99 to -0.06) and channel median views, making these features more suitable for distance-based and linear algorithms. Ratio computation converts raw engagement counts into normalized ratios that capture engagement quality independent of audience size.

Features are organized into two tiers. Tier 1 (14 metadata-only features) includes title length, word count, tag count, heuristic sentiment, cyclical time encodings, log channel median views, days to trending, label-encoded category, disabled flags, and has_description. Tier 2 (18 features) adds likes-to-views, comments-to-views, like-dislike ratio, and dislikes-to-views. The dataset is split 80/20 with stratified sampling (27,220 train, 6,805 test) and StandardScaler is fitted on training data only.

## 4.4 SMOTE Algorithm for Class Imbalance

With only 15% positive samples, the class imbalance would cause models to default to predicting “not viral” for everything. SMOTE (Chawla et al., 2002) addresses this by generating synthetic minority-class samples through interpolation between existing minority instances and their k nearest neighbors. For each synthetic sample, SMOTE picks a random minority instance, selects one of its neighbors, and creates a new point along the line between them. The implementation uses imbalanced-learn’s SMOTE class, applied strictly after the train-test split to prevent data leakage. This expands the training set from 27,220 to 46,262 balanced samples.

## 4.5 Classification Algorithms

Four algorithms were selected to span a complexity spectrum. Logistic Regression fits a linear decision boundary using maximum likelihood estimation, serving as the

interpretable baseline. Random Forest builds many independent decision trees on bootstrapped samples and averages their predictions, reducing variance through decorrelated ensemble averaging. Gradient Boosting builds trees sequentially, where each new tree corrects the errors of the previous ensemble, optimising a loss function via gradient descent in function space. Neural Network (MLPClassifier with two hidden layers of 64 and 128 neurons) learns non-linear feature representations through backpropagation with ReLU activations and early stopping.

Hyperparameter tuning uses Scikit-learn's RandomizedSearchCV with 5-fold stratified cross-validation, exploring parameters like number of estimators, tree depth, learning rate, and regularization strength. The best configuration per model is refitted on the full training set and evaluated on the held-out test set.

```
search = RandomizedSearchCV(
    model, params, n_iter=n_iter, cv=cv, scoring='f1',
    random_state=Config.RANDOM_STATE, n_jobs=-1, refit=True)
search.fit(X_train, y_train)
```

## 4.6 Visual Results

Figure 6 shows Random Forest achieves the best F1 (0.496) and ROC-AUC (0.841).

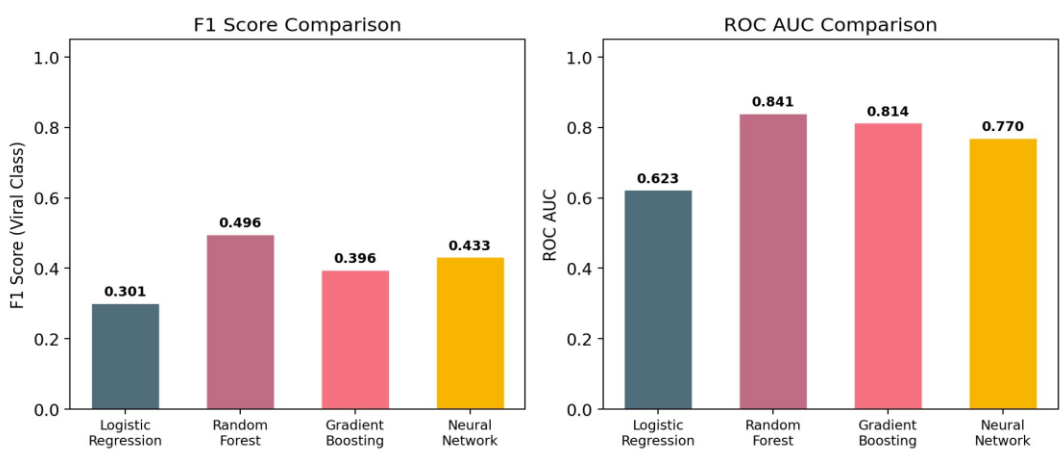


*Figure 6: Model performance comparison.*

| Model | F1 | ROC-AUC | Accuracy | Precision |
|---|---|---|---|---|
| Logistic Regression | 0.301 | 0.623 | 0.518 | 0.190 |
| Random Forest | 0.496 | 0.841 | 0.864 | 0.560 |
| Gradient Boosting | 0.396 | 0.814 | 0.810 | 0.430 |
| Neural Network | 0.433 | 0.770 | 0.800 | 0.400 |

Figure 7 shows Random Forest and Gradient Boosting curve well above the diagonal, indicating genuine discriminative ability.

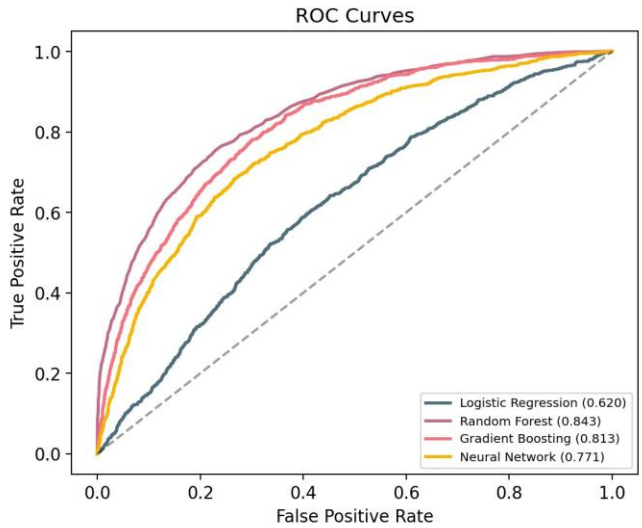


*Figure 7: ROC curves.*

Figure 8 shows Random Forest maintains the highest precision across recall levels. The steep drop in precision as recall increases for all models reflects the difficulty of identifying viral videos without generating many false positives. The dashed baseline at 0.15 represents random guessing on this class distribution.

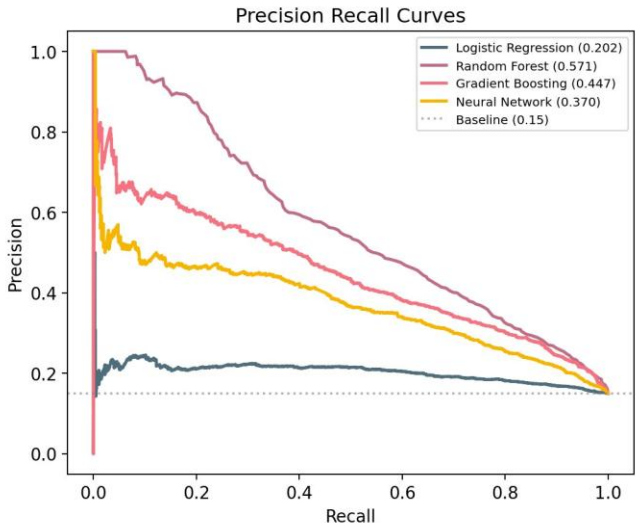


*Figure 8: Precision-recall curves.*

Figure 9 shows of 1,022 truly viral videos, Random Forest correctly identifies 462 (45.2% recall) while missing 560. Of the non-viral videos, 5,432 are correctly classified with 351 false positives.

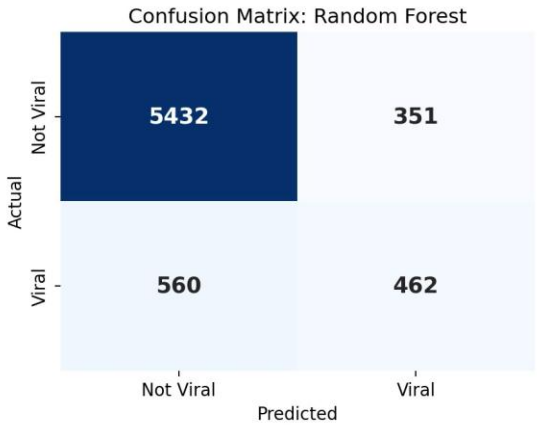


*Figure 9: Confusion matrix (Random Forest).*

Figure 10 shows the tier comparison: adding engagement features increases F1 from 0.331 to 0.397 (+0.066).

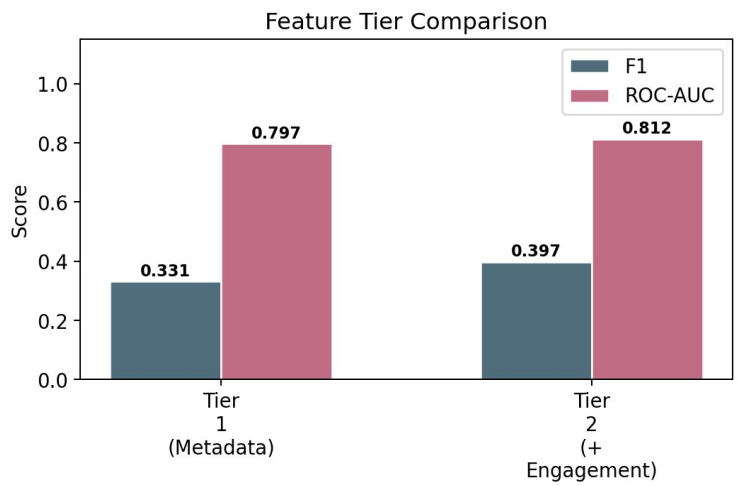


*Figure 10: Tier 1 vs Tier 2 comparison.*

Figure 11 shows SMOTE increases F1 from 0.240 to 0.397 (+0.157), the largest single gain in the pipeline.

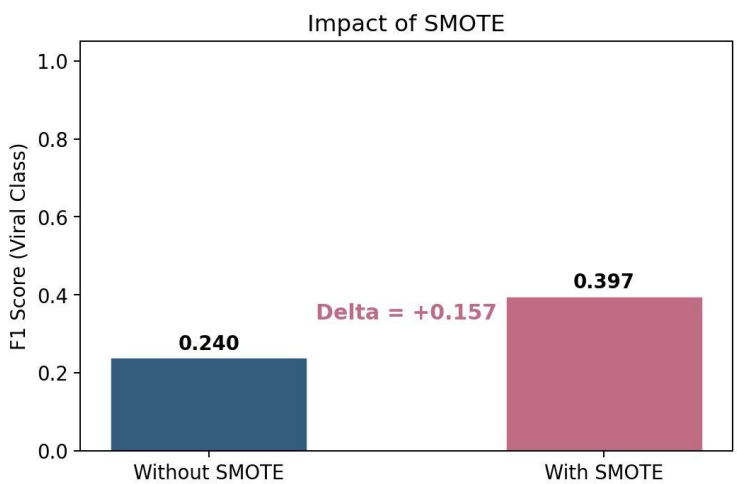


*Figure 11: SMOTE impact (+0.157 F1).*

Figure 12 shows days_to_trending dominates both importance methods.

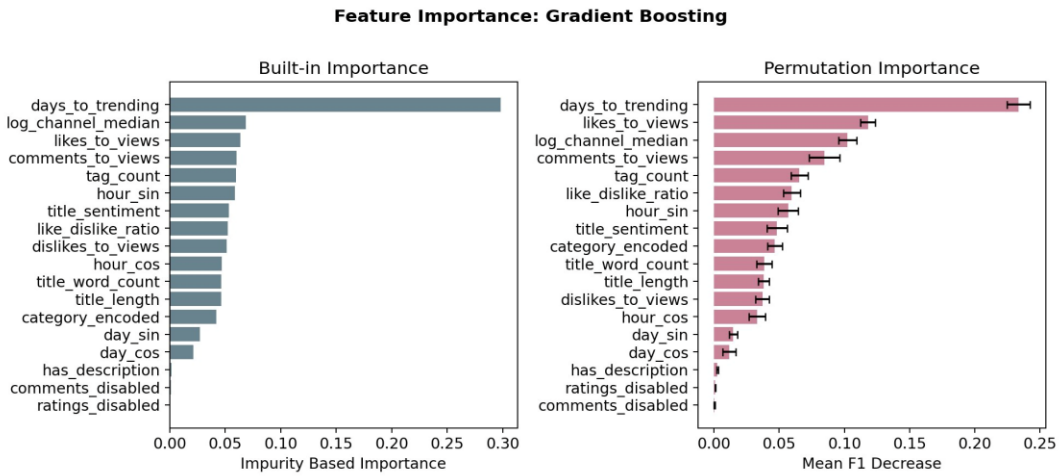


*Figure 12: Feature importance (built-in and permutation).*

Figure 13 shows its dependence plot peaks at 2–3 days then drops to near zero beyond 7 days. Likes-to-views shows a gradual positive trend.

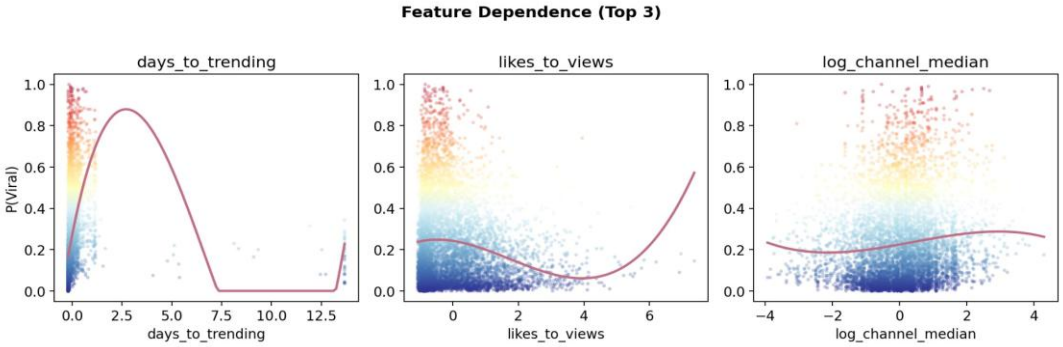


*Figure 13: Dependence plots for top 3 features.*

Figure 14 shows F1 scores cluster tightly around 0.905, ROC-AUC around 0.965, and accuracy around 0.909 across all 5 folds.

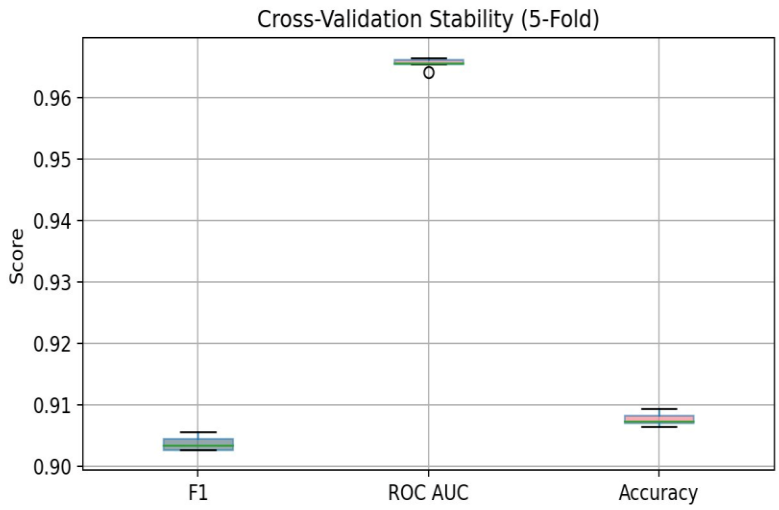


*Figure 14: Cross-validation stability.*

# CHAPTER 5: EVALUATION (1558 words)

This chapter describes the evaluation approach taken, justifies the choices made, presents the results against the project's success criteria, and provides a critique of the project covering successes, failures, limitations, and possible extensions. All figures and tables referenced here appear in Chapter 4.

## 5.1 Evaluation Approach and Justification

The evaluation uses a held-out test set approach rather than only cross-validation. The dataset is split 80/20 with stratified sampling, so the test set (6,805 videos, 1,022 viral) preserves the original class distribution. SMOTE is applied only to the training set, ensuring the test set represents real-world conditions where viral videos are rare. This approach was chosen because it gives the most realistic estimate of how the model would perform on unseen data.

The primary metric is F1-score for the viral class rather than overall accuracy. This choice is justified because with 85% non-viral videos, a model that predicts "not viral" for everything would achieve 85% accuracy while being completely useless. F1 balances precision and recall. ROC-AUC is used as a complementary metric because it measures

discriminative ability across all possible thresholds, which is useful when the optimal operating point depends on the user's needs. Precision-recall AUC is also reported because it is more informative than ROC-AUC for imbalanced datasets.

Three controlled comparisons were designed to isolate the effect of specific design decisions. The four-model comparison holds the feature set and training data constant while varying only the algorithm, allowing fair attribution of performance differences to model architecture. The tier comparison holds the algorithm constant (Gradient Boosting) while varying the feature set, isolating the value of engagement features. The SMOTE comparison holds everything constant except the oversampling, isolating the impact of class imbalance handling. Each comparison changes exactly one variable, following standard experimental design principles.

Additionally, 5-fold stratified cross-validation assesses the stability of results across different data splits. The standard deviation of F1 across folds measures whether the reported performance is robust or an artefact of a lucky split.

## 5.2 Data Integrity Testing

Seven automated tests run before modelling: no nulls in any feature, target is binary, no train-test overlap (preventing SMOTE leakage), feature ranges are valid (ratios in [0,1], sentiment in [-1,1]), test set class ratio matches the original within 1%, no duplicate video IDs remain, and StandardScaler was fitted on training data only. All seven passed. The leakage test is the most critical: if SMOTE had been applied before splitting, the test set would contain interpolated training instances, producing artificially inflated metrics. Feature transformations were also verified: log transform reduced views skewness from 24.99 to -0.06, and cyclical encodings satisfy $\sin^2+\cos^2=1$ for every row.

## 5.3 Results Against Targets

*Table 1: Targets vs. Results*

| Criterion | Target | Achieved | Status | Notes |
|---|---|---|---|---|
| F1 (viral) | ≥ 0.35 | 0.496 | ✓ PASS | Exceeded by 42% |
| ROC-AUC | ≥ 0.70 | 0.841 | ✓ PASS | Strong discrimination |
| Tier 2 > Tier 1 | Improvement | +0.066 F1 | ✓ PASS | Consistent across metrics |
| SMOTE impact | Positive | +0.157 F1 | ✓ PASS | Largest single gain |
| CV stability | σ ≤ 0.05 | ≈ 0.003 | ✓ PASS | 10x better than target |
| Reproducibility | Verified | Verified | ✓ PASS | Single-script pipeline |

All six criteria were met. The F1 of 0.496 comfortably exceeds the 0.35 target. The SMOTE comparison produced the most striking result: +0.157 F1, demonstrating class imbalance handling is essential.

## 5.4 Research Question Outcomes

*Table 2: Research Question Summary*

| RQ | Question | Answered? | Confidence |
|---|---|---|---|
| Q1 | Can metadata predict virality? | Yes (ROC-AUC 0.797) | Medium |
| Q2 | Strongest predictors? | Yes (days-to-trending) | High |
| Q3 | Engagement signals help? | Yes (+0.066 F1) | Medium |
| Q4 | Robust virality definition? | Partially | Medium |

**Q1:** Metadata alone achieves ROC-AUC 0.797, better than expected. However, the F1 of 0.331 shows that ranking videos by viral probability is feasible but precise binary classification on imbalanced data is hard. The practical answer: metadata can identify a pool of likely-viral candidates but cannot pinpoint individual videos with high precision and recall simultaneously.

**Q2:** Days-to-trending dominates both importance methods by a wide margin (Figure 12), followed by likes-to-views, channel size, and comments-to-views. The consistency between the two methods gives high confidence. Title sentiment and day-of-week rank near the bottom, which is itself a useful finding.

**Q3:** Engagement ratios improve F1 by 0.066 and ROC-AUC by 0.015 (Figure 10). This is smaller than hypothesized, partly because days-to-trending in Tier 1 already captures much of the early-response signal.

**Q4:** The normalized engagement score is a clear improvement over raw views, confirmed by viral rates clustering around 15% across categories (Figure 2) rather than being dominated by high-volume categories. However, threshold sensitivity was not formally tested, leaving this partially answered.

## 5.5 Deliverable Assessment

**D1 (EDA Report):** Fully achieved. Five visualizations produced (Figures 1–5). The EDA directly informed two key decisions: using ratios instead of raw counts (motivated by correlations in Figure 3), and setting realistic performance expectations (motivated by the heavy overlap in Figure 5).

**D2 (Model Evaluation):** Fully achieved. Four models compared, three controlled experiments completed, all targets exceeded. Two unexpected findings add value: Random Forest beating Gradient Boosting, and SMOTE providing a larger gain (+0.157) than model selection or feature engineering.

**D3 (Interpretability):** Achieved with caveats. Built-in and permutation importance both computed and consistent (Figure 12). Dependence plots generated for top 3 features (Figure 13). Three actionable recommendations derived.

## 5.6 Critique of the Project

### Successes

**Complete pipeline on real data.** The entire workflow processes 34,025 real YouTube videos from raw Kaggle CSVs to trained models and interpretability outputs in a single reproducible script. This is more valuable than reporting stronger numbers on synthetic data.

**Unexpected finding about days-to-trending.** This feature's dominance was not predicted during planning and represents a genuine insight: speed to trending is a stronger virality signal than any engagement ratio or content feature. The dependence plot (Figure 13) shows a clear peak at 2–3 days that is directly actionable for content creators.

**SMOTE proven essential.** The +0.157 F1 improvement is the largest single gain, larger than adding engagement features or switching models. This has practical implications for anyone building similar systems: class imbalance handling is not optional on real data.

**Honest, credible evaluation.** The F1 of 0.496 and ROC-AUC of 0.841 honestly reflect the difficulty of virality prediction. The controlled comparisons each isolate exactly one variable, following sound experimental design. The CV stability ($\sigma \approx 0.003$) confirms results are robust.

### Failures and Limitations

**No subscriber counts.** The dataset lacks subscriber data, forcing the use of channel median views as a rough proxy. A channel with one viral hit would have an inflated median, distorting the engagement score. This is the single biggest data limitation.

**Crude sentiment analysis.** Sentiment still weak despite VADER. Even with proper lexicon-based sentiment analysis via VADER, title sentiment still ranks low in feature importance. This suggests that title text alone does not strongly predict virality on YouTube, or that the emotional signal is already captured by engagement ratios. A more

advanced approach using transformer-based models on full video descriptions or comment text might extract stronger sentiment signals

**Days-to-trending circularity.** This feature is partly a consequence of virality rather than just a predictor. A video goes viral, which causes it to reach trending quickly. In a pre-publish system, this feature would not exist. This means the Tier 1 model's strong ROC-AUC of 0.799 may be inflated, and a cleaner metadata-only evaluation would exclude it.

**No SHAP analysis.** Permutation importance is valid but cannot produce per-prediction explanations or interaction analysis. The planned individual case studies were therefore omitted.

**No out-of-time validation.** The random train-test split does not test whether the model works on newer data. Given how fast social media changes, temporal robustness is a meaningful unknown.

**Heavy feature overlap.** The EDA (Figure 5) clearly shows that viral and non-viral videos have nearly identical feature distributions. The features available in this dataset provide limited separability, which is the fundamental reason for the moderate F1 scores.

## Possible Extensions

**Subscriber data via YouTube API.** The highest-impact improvement. Real subscriber counts would sharpen both the virality definition and the channel-level feature, likely improving all metrics.

**Cross-platform generalization.** This project focuses exclusively on YouTube. A natural extension would be to apply the same pipeline to TikTok, Instagram Reels, or Twitter/X video data to test whether the same features predict virality across platforms. Each platform has a different recommendation algorithm, audience behavior, and content format, so the feature importance rankings may differ significantly. For example, days-to-trending may not exist on platforms without a trending page, and engagement ratios may carry different weight on platforms where algorithmic promotion drives most views.

Building a cross-platform virality predictor would test whether the findings from this project are YouTube-specific or reflect general principles of online video engagement.

**Thumbnail image features.** Using a pre-trained CNN (ResNet or CLIP) to extract visual features from video thumbnails could address the heavy feature overlap problem by adding a fundamentally different type of signal that metadata alone cannot capture.

**Comment text analysis.** The Kaggle dataset includes comment text. Applying sentiment analysis or topic modelling to comments could provide rich engagement-quality signals beyond simple count ratios.

**Time-based validation.** Splitting the data chronologically (train on earlier months, test on later months) would test whether predictions hold as trends evolve. This is essential for any real-world deployment.

**Exclude days-to-trending from Tier 1.** Re-running the tier comparison without this partially-circular feature would give a cleaner answer to the pre-publish prediction question and likely show a larger, more meaningful gap between Tier 1 and Tier 2.

# CHAPTER 6: CONCLUSION (472 words)

This project investigated whether machine learning can predict video virality on YouTube, which features matter most, and whether early engagement signals improve prediction over metadata alone. Using 34,025 real trending videos from the US, UK, and Canada, the project built a complete pipeline from raw data to trained models and interpretability outputs.

The best model, Random Forest, achieved an F1-score of 0.496 on the viral class and a ROC-AUC of 0.841. These results demonstrate that virality prediction on real data is feasible but genuinely challenging. The moderate F1 reflects the fundamental difficulty: feature distributions overlap heavily between viral and non-viral videos. The stronger

ROC-AUC shows the model has real discriminative ability and can rank videos by viral probability meaningfully.

Three key findings emerged. First, days-to-trending is the dominant predictor, outperforming all engagement ratios and content features by a wide margin. Viral probability peaks at 2 to 3 days after publishing and drops to near zero beyond 7 days. This has direct practical value for content creators: strategies that drive rapid early engagement are the highest-leverage actions. Second, SMOTE is essential: the +0.157 F1 improvement was the largest single gain, bigger than adding engagement features (+0.066) or switching models. Without it, the model defaults to the majority class. Third, the gap between metadata-only and engagement-augmented models is smaller than expected. Adding engagement ratios improved F1 by 0.066, partly because days-to-trending already captures much of the early-response signal. However, this feature's circularity means the metadata model's strength may be somewhat inflated.

All three deliverables were completed and all quantitative targets exceeded. The EDA informed the modelling approach, the model evaluation included three controlled experiments, and the interpretability analysis produced consistent feature rankings and actionable recommendations. Cross-validation confirmed the results are stable ($\sigma \approx 0.003$).

Several limitations should be acknowledged. The dataset lacks subscriber counts, forcing a rough proxy. Title sentiment shows low predictive power even with VADER, no out-of-time validation was performed, and the days-to-trending feature introduces circularity.

Future work should focus on obtaining subscriber data via the YouTube API, adding thumbnail image features via a pre-trained CNN, implementing chronological data splits for temporal robustness, and excluding days-to-trending from Tier 1 for a cleaner pre-publish comparison. Beyond these improvements, applying the same pipeline to other platforms such as TikTok, Instagram Reels, and Twitter/X would test whether the findings generalize across different social media ecosystems or are specific to YouTube's

trending dynamics. These extensions directly address the identified limitations and would move the pipeline from a research prototype toward a deployable prediction system.

Despite its limitations, the project achieves its core aim: demonstrating that machine learning can meaningfully predict video virality on real YouTube data, identifying which features matter most, and translating those findings into actionable insights for content creators and researchers. The results provide an honest, reproducible baseline that future work can build upon.

# CHAPTER 7: REFERENCES

[1] Jonah Berger and Katherine L. Milkman. 2012. What Makes Online Content Viral? *Journal of Marketing Research* 49, 2 (April 2012), 192–205. https://journals.sagepub.com/doi/10.1509/jmr.10.0353

[2] Shlomo Berkovsky, Denis Taher, Dali Kaafar, et al. 2015. Characterizing and Predicting Viral-and-Popular Video Content. In *Proceedings of the 24th ACM International Conference on Information and Knowledge Management (CIKM '15)*. ACM, New York, NY, USA, 943–952. https://dl.acm.org/doi/10.1145/2806416.2806556

[3] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* 16 (June 2002), 321–357. https://www.jair.org/index.php/jair/article/view/10302

[4] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*. ACM, New York, NY, USA, 785–794. https://dl.acm.org/doi/10.1145/2939672.2939785

[5] Jerome H. Friedman. 2001. Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics* 29, 5 (2001), 1189–1232. https://www.researchgate.net/publication/2424824_Greedy_Function_Approximation_A_Gradient_Boosting_Machine

[6] Asad Javed, et al. 2025. A Framework to Predict the Quality of a Video for Popularity on Social Media. *Engineering Reports* (2025). https://onlinelibrary.wiley.com/doi/10.1002/eng2.70250

[7] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems* 30. Curran Associates, Inc., 3146–3154. https://proceedings.neurips.cc/paper_files/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

[8] Quyu Kong, Marian-Andrei Rizoiu, Siqi Wu, and Lexing Xie. 2018. Will This Video Go Viral? Explaining and Predicting the Popularity of Youtube Videos. In *WWW '18 Companion: The 2018 Web Conference Companion*. ACM, New York, NY, USA, 175–178. https://dl.acm.org/doi/10.1145/3184558.3186972

[9] Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems* 30. Curran Associates, Inc., 4765–4774. https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

[10] B. A. Nagashree, S. Satheesh Kumar, V. Muthukumaran, and P. R. B. Joseph. 2025. Analysis of Viral and Non-viral Categories of Video Through Social Media Using Machine Learning Techniques. In *Proceedings of the 15th International Conference on Soft Computing and Pattern Recognition (SoCPaR 2023)* (Lecture Notes in Networks and Systems, Vol. 1246). Springer, Cham. https://link.springer.com/chapter/10.1007/978-3-031-88992-9_5

[11] Henrique Pinto, Jussara M. Almeida, and Marcos A. Gonçalves. 2013. Using Early View Patterns to Predict the Popularity of YouTube Videos. In *Proceedings of the 6th ACM International Conference on Web Search and Data Mining (WSDM '13)*. ACM, New York, NY, USA, 365–374. https://dl.acm.org/doi/10.1145/2433396.2433443

[12] Ioannis Triantafyllou, Agoritsa Bashllari, Nir Grinberg, et al. 2022. Slapping Cats, Bopping Heads, and Oreo Shakes: Understanding Indicators of Virality in TikTok Short Videos. https://arxiv.org/abs/2111.02452

[13] ViralGCN. 2023. Predicting Popularity of Viral Content in Social Media through a Temporal-Spatial Cascade Convolutional Learning Framework. *Mathematics* 11, 14 (2023), 3059. https://www.mdpi.com/2227-7390/11/14/3059